

```
[1] 0.101291
```

Example 3 - ppois

Consider that the number of visits on a web page is known to follow a Poisson distribution with mean 15 visits per hour. What is the probability of getting

1. 10 or less visits per hour?
2. The probability of getting more than 20 visits per hour
3. The probability of getting 10 to 20 visits per hour.

Solution

```
ppois(10, lambda = 15)
```

```
[1] 0.1184644
```

```
ppois(20, lambda = 15, lower.tail = FALSE)
```

```
[1] 0.08297091
```

```
ppois(20, lambda = 15) - ppois(10, lambda = 15)
```

```
[1] 0.7985647
```

Continuous Probability Distributions

Exponential Distribution

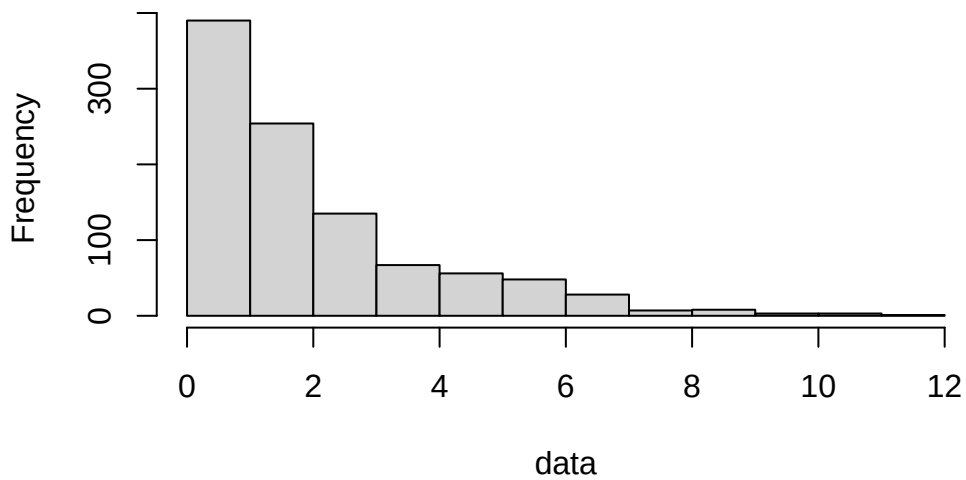
The pdf is given as:

$$f(x) = \lambda e^{-\lambda x}$$

To simulate observations using exponential distribution use the function `rexp(n, rate)`. The rate is given as $rate = \frac{1}{\lambda}$

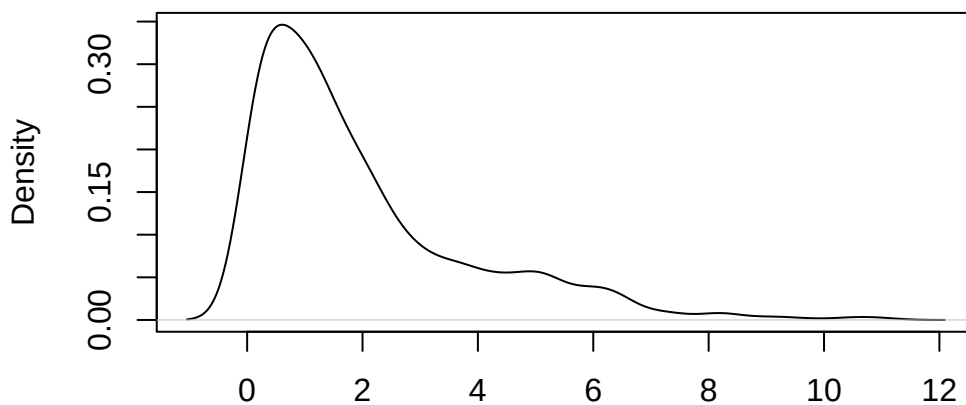
```
data <- rexp(1000, rate = 0.5)
hist(data)
```

Histogram of data



```
plot(density(data))
```

density(x = data)



N = 1000 Bandwidth = 0.3481

The function in R that is used to calculate the density function at any rate λ is the `dexp()` function

```
# Grid of values
x <- seq(from = 0, to = 8, by = 0.01)
# Exponential PDF of rate 2
dexp(x, rate = 2)
```

The R function that allows you to calculate the probabilities of a random variable XX taking values lower than x is the `pexp` function

Example 1

The time spent on a determined web page is known to have an exponential distribution with an average of 5 minutes per visit. To calculate the probability of a visitor spending up to 3 minutes on the site

```
pexp(3, rate = 0.2)
```

```
[1] 0.4511884
```

Example 2

The time (in hours) required to repair a machine is an exponential distributed random variable with paramter $\lambda = \frac{1}{2}$

1. Find the value of the density function at $x=2.5$.
2. Find the probability that a repair time takes at most 3 hours.
3. Find the probability that a repair time exceeds 4 hours.
4. Find the probability that a repair time takes between 2 to 4 hours.
5. Find the 50th quantile of given Exponential distribution.
6. Simulate 1000 Exponential distributed random variables with $\lambda = \frac{1}{2}$

Solution

```
# 1
dexp(2.5,rate=0.5)
# 2
pexp(3,rate=0.5)
# 3
pexp(4,rate=0.5,lower.tail=FALSE)
# 4
pexp(4,rate=lambda)- pexp(2,rate=0.5)
# 5
qexp(0.50,rate=0.5).
# 6
rexp(1000,0.5)
```

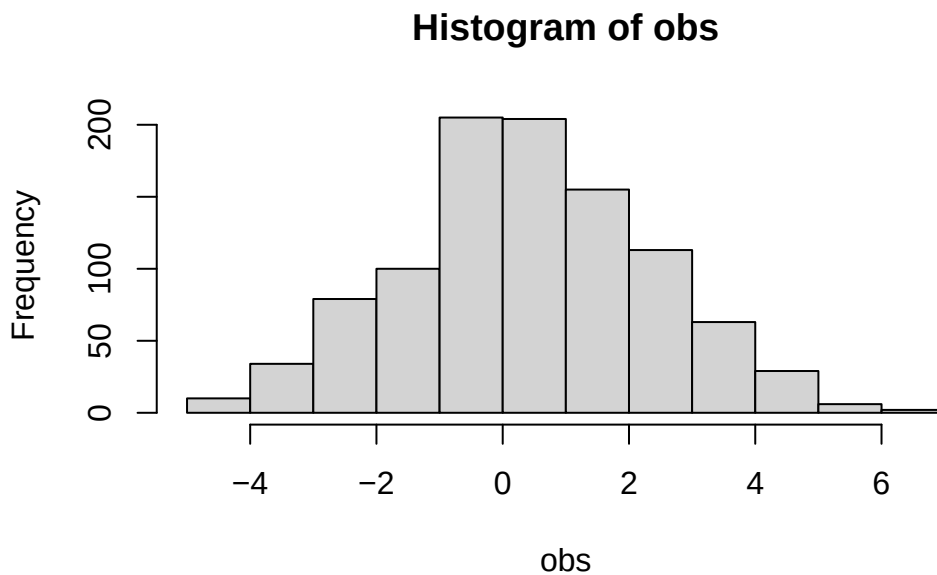
Normal Distribution

The normal distribution is given as:

$$f(X = x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

To simulate observations from a normal distribution use the `rnorm(n,mean,sd)`

```
obs <- rnorm(1000,0.4,2)
hist(obs)
```



The function `dnorm` returns the value of the probability density function (pdf) of the normal distribution given a certain random variable x .

The function `pnorm` returns the value of the cumulative density function (cdf) of the normal distribution given a certain random variable q .

Example 1

Suppose the height of males at a certain school is normally distributed with a mean of $\mu=70$ inches and a standard deviation of $\sigma = 2$ inches. Approximately what percentage of males at this school are taller than 74 inches?

```
pnorm(74, mean=70, sd=2, lower.tail=FALSE)
```

```
[1] 0.02275013
```

Example 2

Suppose the height of plants in a certain region is normally distributed with a mean of $\mu=13$ inches and a standard deviation of $\sigma = 2$ inches. Approximately what percentage of plants in this region are between 10 and 14 inches tall?

```
pnorm(14, mean=13, sd=2) - pnorm(10, mean=13, sd=2)
```

```
[1] 0.6246553
```

Example 3

Suppose that you have a machine that packages rice inside boxes. The process follows a Normal distribution and it is known that the mean of the weight of each box is 1000 grams and the standard deviation is 10 grams.

What is the probability of a box weighing exactly 950 grams?

```
dnorm( 950, 1000, 10)
```

```
[1] 1.48672e-07
```

What is the probability of a box weighing more than 980 grams?

```
pnorm(980,1000,10,lower.tail=FALSE)
```

```
[1] 0.9772499
```

Uniform Distribution

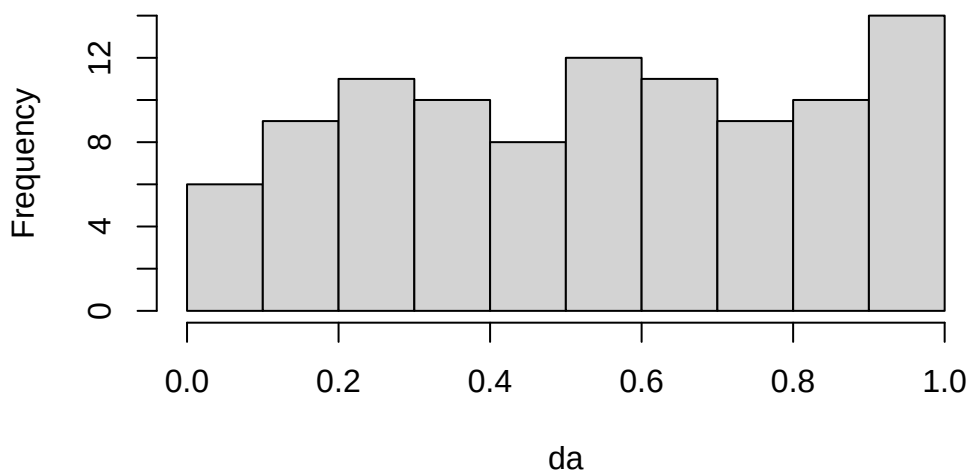
A uniform distribution, also called a rectangular distribution, is a probability distribution that has constant probability.

This distribution is defined by two parameters, a and b:

To simulate observations from the uniform distribution use `runif()` and specify the minimum and maximum

```
da <- runif(100, min = 0, max = 1)
hist(da)
```

Histogram of da



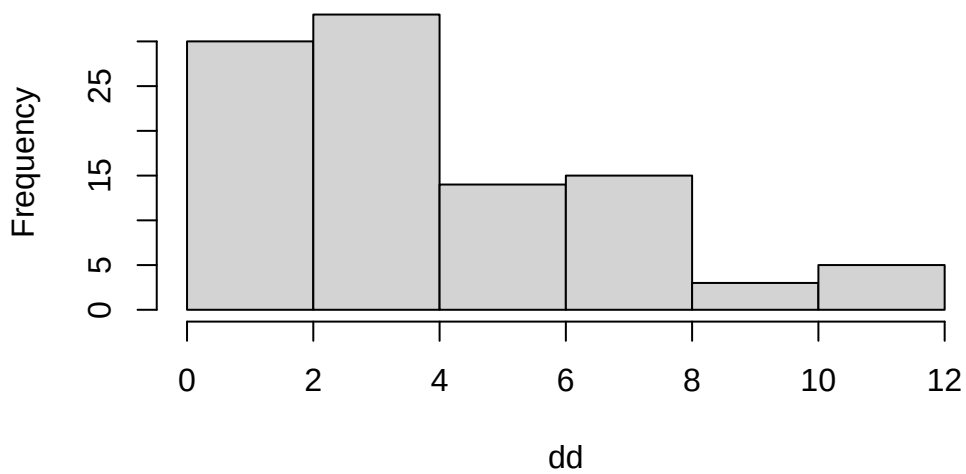
The other functions are the `punif()`, `qunif()` and `dunif()` functioning as the previous distribution.

Chi-square Distribution

Chi-square distribution is characterized by the degrees of freedom

```
dd <- rchisq(100, df = 4)
hist(dd)
```

Histogram of dd



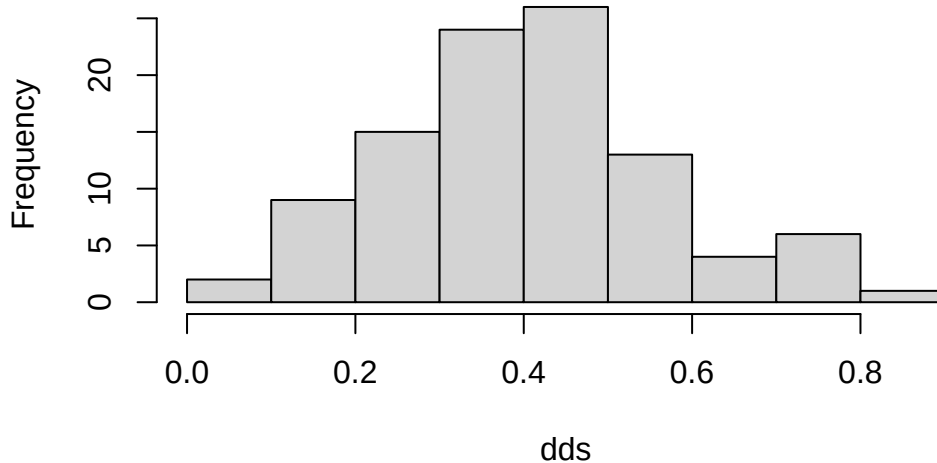
The other functions are the `pchisq()`, `qchisq()` and `dchisq()` functioning as the previous distribution.

Beta Distribution

Beta distribution is characterized by the shape and scale parameters

```
dds <- rbeta(100, shape1 = 3, shape2 = 4)
hist(dds)
```

Histogram of dds



The other functions are the `pbeta()`, `qbeta()` and `dbeta()` functioning as the previous distribution.